

Result clarification

The contrasting performance likely comes from the interplay between the embedding spaces produced by the two models and the assumptions of the classifiers.

BERT embeddings may structure semantic information in a way that naturally forms clusters that are easier to separate linearly. This makes them suitable for linear classifiers like logistic regression, which can then easily find clear decision boundaries. On the other hand, embeddings from models like all-MiniLM-L6-v2 might represent semantic relationships in a more complex way, making it harder for linear methods alone to capture meaningful patterns. In these cases, more flexible, non-linear approaches might be needed to effectively classify and interpret the data.

Logistic Regression is inherently linear, meaning it can only capture patterns that fit neatly along linear boundaries. So, if BERT embeddings naturally align with these simpler, linear patterns, logistic regression tends to perform well. In contrast, Random Forest, which combines multiple decision trees, excels at capturing more complex, non-linear relationships. Its superior performance on embeddings from all-MiniLM-L6-v2 suggests that these embeddings likely contain richer, non-linear structures—structures that logistic regression alone cannot fully grasp.